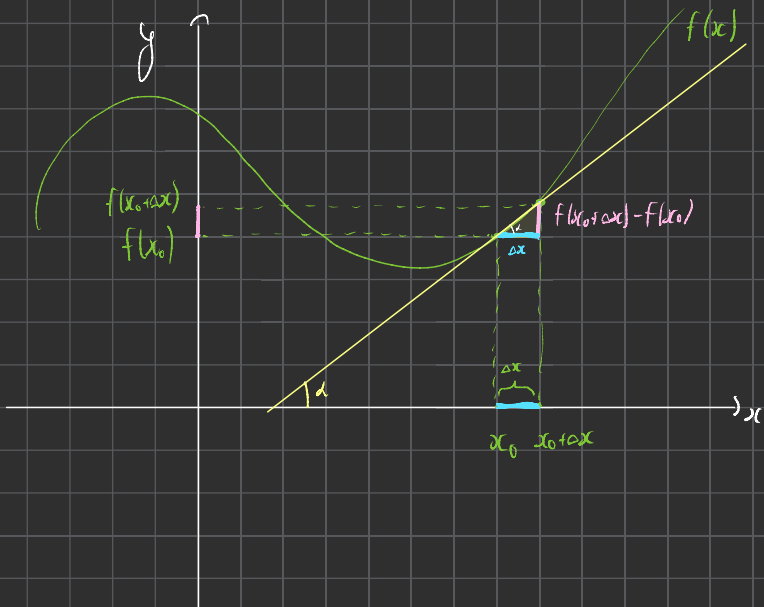


Вед №10 Производная



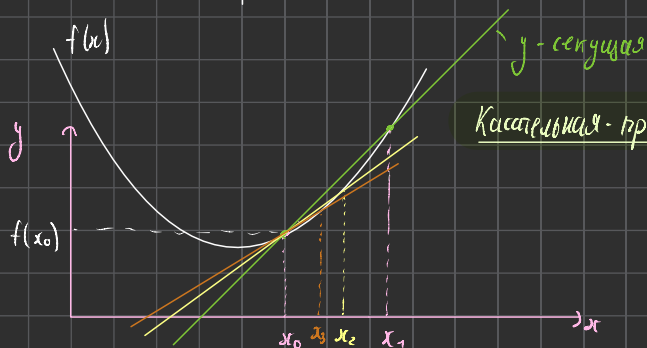
Плоск $f(x)$ от θ $U_{\delta}(x_0)$

$$\text{Тогда } f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} =$$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

$$\text{т.е. } \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = f'(x_0)$$

т.е. Δ - угол наклона касательной к $f(x)$ в x_0



Касательная - предельное положение секущих

Покажем, что если $f(x)$ имеет произв в т. x_0 , то она непр. в этой точке:

$$\square f(x) - \text{гип в } x_0 \quad \exists f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = A \quad \hookrightarrow \quad \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = A + \overset{0}{\alpha(\Delta x)}$$

$$f(x_0 + \Delta x) - f(x_0) = A \Delta x + \alpha(\Delta x) \Delta x$$

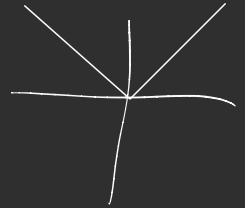
$$f(x_0 + \Delta x) = f(x_0) + A \Delta x + \alpha(\Delta x) \Delta x$$

$$\lim_{\Delta x \rightarrow 0} f(x_0 + \Delta x) = \lim_{\Delta x \rightarrow 0} \left(f(x_0) + \underbrace{A \Delta x}_{\downarrow 0} + \underbrace{\alpha(\Delta x) \Delta x}_{\substack{\downarrow 0 \\ \downarrow 0}} \right) = f(x_0) - \text{функция непр в т. } x_0 \quad \square$$

Дифференцируемость $\xrightarrow{\quad}$ Непрерывность



$y = |x|$ - непр, но не гип в т. $x_0 = 0$



Свойства Производных

Пусть $f(x)$ и $g(x)$ имеют конечную произв. в точке x_0 , тогда.

$$1) \left(f(x) + g(x) \right)' \Big|_{x=x_0} = f'(x_0) + g'(x_0)$$

$$2) \left(f(x) \cdot g(x) \right)' \Big|_{x=x_0} = f'(x_0) \cdot g(x_0) + f(x_0) \cdot g'(x_0)$$

$$3) \left(\frac{f(x)}{g(x)} \right)' \Big|_{x=x_0} = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g^2(x_0)}, \quad g(x_0) \neq 0$$

$$\boxed{1} \quad \left(f(x) + g(x) \right)' \Big|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{(f(x_0 + \Delta x) + g(x_0 + \Delta x)) - (f(x_0) + g(x_0))}{\Delta x} =$$

$$= \lim_{\Delta x \rightarrow 0} \frac{(f(x_0 + \Delta x) - f(x_0)) + (g(x_0 + \Delta x) - g(x_0))}{\Delta x} = \lim_{\Delta x \rightarrow 0} \left(\frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} + \frac{g(x_0 + \Delta x) - g(x_0)}{\Delta x} \right) =$$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} + \lim_{\Delta x \rightarrow 0} \frac{g(x_0 + \Delta x) - g(x_0)}{\Delta x} = f'(x_0) + g'(x_0) \quad \square$$

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$
$$g'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{g(x_0 + \Delta x) - g(x_0)}{\Delta x}$$

2

$$\left(f(x) \cdot g(x) \right)' \Big|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) g(x_0 + \Delta x) - f(x_0) g(x_0)}{\Delta x} =$$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) g(x_0 + \Delta x) - f(x_0) g(x_0) + f(x_0 + \Delta x) g(x_0) - f(x_0 + \Delta x) g(x_0)}{\Delta x} =$$

$$= \lim_{\Delta x \rightarrow 0} \frac{g(x_0) (f(x_0 + \Delta x) - f(x_0)) + f(x_0 + \Delta x) (g(x_0 + \Delta x) - g(x_0))}{\Delta x} =$$

$$= \lim_{\Delta x \rightarrow 0} g(x_0) \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} + \lim_{\Delta x \rightarrow 0} \overbrace{f(x_0 + \Delta x)}^{f(x_0)} \cdot \frac{g(x_0 + \Delta x) - g(x_0)}{\Delta x} = f'(x_0) g(x_0) + f(x_0) g'(x_0) \quad \square$$

3

$$\left(\frac{f(x)}{g(x)} \right)' \Big|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{\frac{f(x_0 + \Delta x)}{g(x_0 + \Delta x)} - \frac{f(x_0)}{g(x_0)}}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) g(x_0) - f(x_0) g(x_0 + \Delta x)}{\Delta x g(x_0 + \Delta x) g(x_0)} =$$

$$= \frac{\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) g(x_0) - f(x_0) g(x_0 + \Delta x)}{\Delta x}}{\lim_{\Delta x \rightarrow 0} g(x_0 + \Delta x) g(x_0)} = \frac{\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) g(x_0) - f(x_0) g(x_0 + \Delta x)}{\Delta x}}{g'(x_0)} \quad \textcircled{=}$$

$$\textcircled{=} \frac{\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) g(x_0) - f(x_0) g(x_0 + \Delta x) + f(x_0) g(x_0) - f(x_0) g(x_0)}{\Delta x}}{g'(x_0)} = \frac{\lim_{\Delta x \rightarrow 0} \frac{g(x_0) (f(x_0 + \Delta x) - f(x_0)) - f(x_0) (g(x_0 + \Delta x) - g(x_0))}{\Delta x}}{g'(x_0)} = \frac{f'(x_0) g(x_0) - f(x_0) g'(x_0)}{g^2(x_0)} \quad \square$$

f(x_0)

// - next p.

g'(x_0)

$$\parallel \lim_{\Delta x \rightarrow 0} f(x_0 + \Delta x) \cdot \frac{g(x_0 + \Delta x) - g(x_0)}{\Delta x} = \boxed{\lim_{\Delta x \rightarrow 0} f(x_0 + \Delta x)} \cdot \boxed{\lim_{\Delta x \rightarrow 0} \frac{g(x_0 + \Delta x) - g(x_0)}{\Delta x}} \parallel$$

Табличные Производные.

1) $f(x) = C$ - константа

$$f(x_0) = C$$

$$f(x_0 + \Delta x) = C$$

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\overset{0}{\cancel{C-C}}}{\underset{0}{\Delta x}} \neq \left[\frac{0}{0} \right] = 0$$

2) $f(x) = x$

$$f(x_0) = x_0$$

$$f(x_0 + \Delta x) = x_0 + \Delta x$$

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\overset{1}{x_0 + \Delta x} - x_0}{\Delta x} = 1$$

3) $f(x) = x^2$

$$f(x_0) = x_0^2$$

$$f(x_0 + \Delta x) = (x_0 + \Delta x)^2$$

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{x_0^2 + 2x_0\Delta x + \overset{0}{\Delta x^2} - x_0^2}{\Delta x} = \lim_{\Delta x \rightarrow 0} (2x_0 + \Delta x) = \textcircled{2x_0}$$

$$f(x) = x^2 + 3x + 1$$

$$f(2) = 2^2 + 3 \cdot 2 + 1$$

$$f(y) = y^2 + 3y + 1$$

$$f(x_0 + \Delta x) = (x_0 + \Delta x)^2 + 3(x_0 + \Delta x) + 1$$

$$4) f(x) = e^x$$

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{e^{x_0 + \Delta x} - e^{x_0}}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{e^{x_0} (e^{\Delta x} - 1)}{\Delta x} = e^{x_0} \lim_{\Delta x \rightarrow 0} \frac{e^{\Delta x} - 1}{\Delta x} = e^{x_0}$$

$$5) f(x) = a^x$$

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{a^{x_0 + \Delta x} - a^{x_0}}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{a^{x_0} (a^{\Delta x} - 1)}{\Delta x} = a^{x_0} \lim_{\Delta x \rightarrow 0} \frac{a^{\Delta x} - 1}{\Delta x} = a^{x_0} \ln a$$

$$6) f(x) = \ln x$$

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\ln(x_0 + \Delta x) - \ln x_0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\ln \frac{x_0 + \Delta x}{x_0}}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\ln \left(1 + \frac{\Delta x}{x_0}\right)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\frac{\Delta x}{x_0}}{\Delta x} = \frac{1}{x_0}$$

$$7) f(x) = \sin x$$

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\sin(x_0 + \Delta x) - \sin x_0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\sin x_0 \cos \Delta x + \sin \Delta x \cos x_0 - \sin x_0}{\Delta x} =$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\sin x_0 \left(\cos \Delta x - 1\right) + \sin \Delta x \cos x_0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \left[\sin x_0 \cdot \frac{-\Delta x^2}{2} + \cos x_0 \right] = \cos x_0$$

Gronce:

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\sin(x_0 + \Delta x) - \sin x_0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{2 \sin \frac{\Delta x}{2} \cos \frac{x_0 + \Delta x}{2}}{\Delta x} = \lim_{\Delta x \rightarrow 0} \left(\frac{\sin \frac{\Delta x}{2}}{\frac{\Delta x}{2}} \right) \cos \frac{x_0 + \Delta x}{2} = \cos x_0$$

$$8) f(x) = \cos x$$

$$f'(x_0) = Df = -\sin x_0$$

$$9) f(x) = \operatorname{tg} x$$

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

$$f'(x_0) = \left(\frac{\sin x}{\cos x}\right)' \Big|_{x=x_0} = \frac{(\sin x)' \Big|_{x_0} \cos x_0 - \sin x_0 (\cos x)' \Big|_{x_0}}{\cos^2 x_0} = \frac{\cos^2 x_0 + \sin^2 x_0}{\cos^2 x_0} = \frac{1}{\cos^2 x_0}$$

$$10) f(x) = \operatorname{ctg} x$$

$$f'(x_0) = Df = -\frac{1}{\sin^2 x_0}$$

$$11) f(x) = x^a$$

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{(x_0 + \Delta x)^a - x_0^a}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{x_0^a \left(1 + \frac{\Delta x}{x_0}\right)^a - x_0^a}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{x_0^a \left(\sqrt[a]{1 + \frac{\Delta x}{x_0}} - 1\right)}{\Delta x} = \lim_{\Delta x \rightarrow 0} x_0^a a \frac{1}{x_0} = a x_0^{a-1}$$

Таблица Производных

$f(x)$	$f'(x)$
c, const	0
$a \cdot x^{\alpha-1}$	$a \cdot x^{\alpha-1}$
a^x	$a^x \ln a$
e^x	e^x
$\log_a x$	$\frac{1}{x \ln a}$
$\ln x$	$\frac{1}{x}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\frac{1}{\cos^2 x}$
$\cot x$	$-\frac{1}{\sin^2 x}$
$\operatorname{arcsin} x$	$\frac{1}{\sqrt{1-x^2}}$
$\operatorname{arctan} x$	$\frac{1}{1+x^2}$
$\operatorname{arccot} x$	$-\frac{1}{1+x^2}$
$\operatorname{arcsinh} x$	$\frac{1}{\sqrt{1+x^2}}$
$\operatorname{arcosh} x$	$\frac{1}{\sqrt{x^2-1}}$

на пальцах

дублирует в 5-10% ВУЗ-ов

Свойства Производных

$u(x)$ и $v(x)$ - гур, C - константа

$$(u+v)' = u' + v'$$

$$(u-v)' = u' - v'$$

$$(C \cdot u)' = C \cdot u'$$

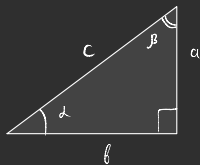
$$(u \cdot v)' = u'v + uv'$$

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

Произв. обр. функций

$$y = \arcsin x; x = \sin y; x' = \cos y = \frac{dx}{dy}$$

$$y' = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - \sin^2 y}} = \frac{1}{\sqrt{1 - x^2}}$$



$$\sin \alpha = \frac{a}{c}$$

$$\alpha = \arcsin \frac{a}{c}$$

$$\cos \beta = \frac{a}{c}$$

$$\beta = \arccos \frac{a}{c}$$

$$\frac{\pi}{2}$$

$$\alpha + \beta = \arcsin \left(\frac{a}{c} \right) + \arccos \left(\frac{a}{c} \right) \Rightarrow \arcsin x + \arccos x = \frac{\pi}{2}$$

$$\arccos x = \frac{\pi}{2} - \arcsin x$$

const

$$(\arccos x)' = \left(\frac{\pi}{2} - \arcsin x \right)' = \left(\frac{\pi}{2} \right)' - (\arcsin x)' = 0 - \frac{1}{\sqrt{1 - x^2}} = -\frac{1}{\sqrt{1 - x^2}}$$

$$y = \arctg x; x = \tg y; x' = \frac{1}{\cos^2 y} = \frac{dx}{dy}$$

$$y' = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{\frac{\cos^2 y + \sin^2 y}{\cos^2 y}} = \frac{1}{1 + \tg^2 y} = \frac{1}{1 + x^2}$$

$$y = \arctg x - DB$$

$$// y = \ln x \rightarrow x = e^y \rightarrow x' = e^y = \frac{dx}{dy}$$

$$y' = \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{e^y} = \frac{1}{x} //$$

$f(x)$	$f'(x)$
$c^{\text{конст}}$	0
$x^{\text{степень}}$	$a \cdot x^{a-1}$
$a^{\text{логарифм}}$	$a^x \ln a$
e^x	e^x
$\log_a x^{\text{логарифм}}$	$\frac{1}{x \ln a}$
$\ln x$	$\frac{1}{x}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\frac{1}{\cos^2 x}$
$\cot x$	$-\frac{1}{\sin^2 x}$
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$
$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$
$\arctan x$	$\frac{1}{1+x^2}$
$\operatorname{arccot} x$	$-\frac{1}{1+x^2}$

// $x = x'$; $x' = (x'')' = 1 \cdot x'' = 1$ // Практика

$$1) (x^3 + 2x^2 - 3x + 7)' = (x')' + (2x^2)' - (3x)' + (7)' = 3x^2 + 2 \cdot 2x - 3 \cdot 1 + 0$$

$$2) (\sin 2x)' = (2 \sin x \cos x)' = 2(\sin x \cos x)' = 2(\underbrace{\sin' x \cos x}_{\cos^2 x} + \underbrace{\sin x \cos' x}_{-\sin^2 x}) = 2(\cos^2 x - \sin^2 x) = 2 \cos 2x$$

$$3) (\tan x \cdot \cot x)' = \left(\frac{\sin x}{\cos x} \cdot \frac{\cos x}{\sin x} \right)' = (1)' = 0$$

$$4) (\cos^2 x + \sin^2 x)' = (1)' = 0$$

$$5) (\ln 3)' = 0$$

$$6) \left(\frac{\sqrt[3]{\pi} \ln 10^5 - \sin 15^\circ}{\cos 30^\circ} \right)' = (\text{конст})' = 0$$

$$7) (\sqrt{x})' = (x^{\frac{1}{2}})' = \frac{1}{2} x^{\frac{1}{2}-1} = \frac{1}{2} x^{-\frac{1}{2}}$$

$$8) \left(\frac{1}{x} \right)' = \frac{1'x - 1x'}{x^2} = \frac{0 \cdot 1 - 1 \cdot x'}{x^2} = -\frac{1}{x^2} \text{ - не нужно так}$$

$$\text{или } (x^{-1})' = -1 x^{-1-1} = -\frac{1}{x^2}$$

$$9) (\ln 3x)' = (\ln 3 + \ln x)' = 0 + \frac{1}{x} = \frac{1}{x}$$

$$10) (\ln x^5)' = (5 \ln x)' = 5(\ln x)' = \frac{5}{x}$$

$$11) (\sqrt[4]{x^3})' = (x^{\frac{3}{4}})' = \frac{3}{4} x^{\frac{3}{4}-1} = \frac{3}{4} x^{-\frac{1}{4}}$$

$$12) (5^{-x})' = (5^{\frac{1}{-x}})' = \left(5^{\frac{1}{u}} \right)' = \left(\frac{5}{u} \right)' = \sqrt[5]{5} \ln \frac{1}{5}$$

$$\cdot \ln x + \ln y = \ln(xy)$$

$$\cdot \ln x - \ln y = \ln \frac{x}{y}$$

$$\cdot \ln x^a = a \ln x$$

$$x^{a+b} = x^a \cdot x^b$$

$$x^{a-b} = \frac{x^a}{x^b}$$

$$(x^a)^b = x^{a \cdot b}$$

$$a^x \cdot b^x = (ab)^x$$

$$\frac{a^x}{b^x} = \left(\frac{a}{b} \right)^x$$